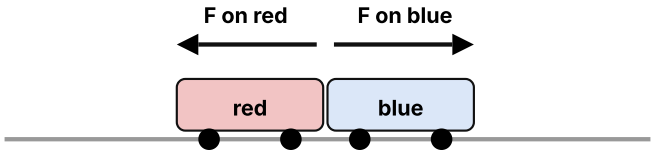
Mini-Lesson 5: Conservation of Momentum

AP Physics 1 · Unit 4 · Do with the mini-lesson before class Thu 11/19. Fill in each box when the lesson shows its orange tag.

Scan with your iPad camera to open the lesson

Name _____ Period _____

A Why the total doesn't change

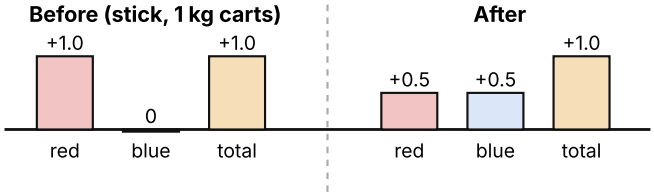


3rd law: the forces are _____. They last the same _____.

So the impulses are _____, and Δp_{red} = _____.

The **total** momentum _____.

B Total before = total after



_____ (Σ means _____)

True for an **isolated system**: no _____.

Stick: _____ Bounce: _____

Start together, fly apart: _____

C The 5-step method

1. _____ (before and after, pick +)

2. Write _____ ← earns a point on the AP exam

3. _____. If they stick, _____.

4. Change p into _____

5. _____

D Worked example: the tackle

Before



After



(80)(5.5) + (88)(_____) = (_____)v′

v′ = _____, moving in _____

E Check questions: draw before/after and show your work

1. Stick: 2.0 kg at +3.0 m/s hits 1.0 kg at rest.

work:

v′ = _____

2. Explode: 60 kg and 40 kg skaters push apart; 60 kg moves −2.0 m/s.

work:

v = _____

3. Bounce: 0.50 kg at +4.0 m/s hits 1.5 kg at rest, bounces back at −2.0 m/s.

work:

v = _____

Big idea in one sentence:

_____

_____